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NLP and Word Embeddings

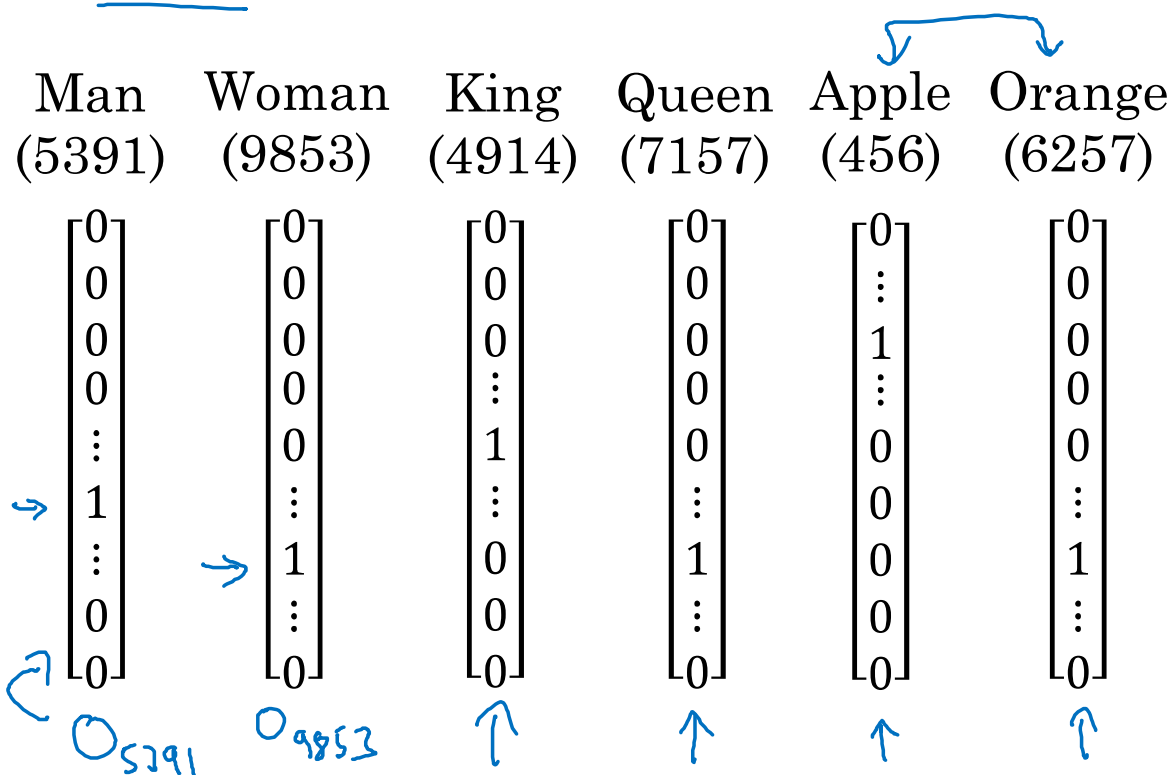
Word representation

Word representation

$V = [a, aaron, \dots, zulu, <UNK>]$

$|V| = 10,000$

1-hot representation



I want a glass of orange juice.

I want a glass of apple _____.

Featurized representation: word embedding

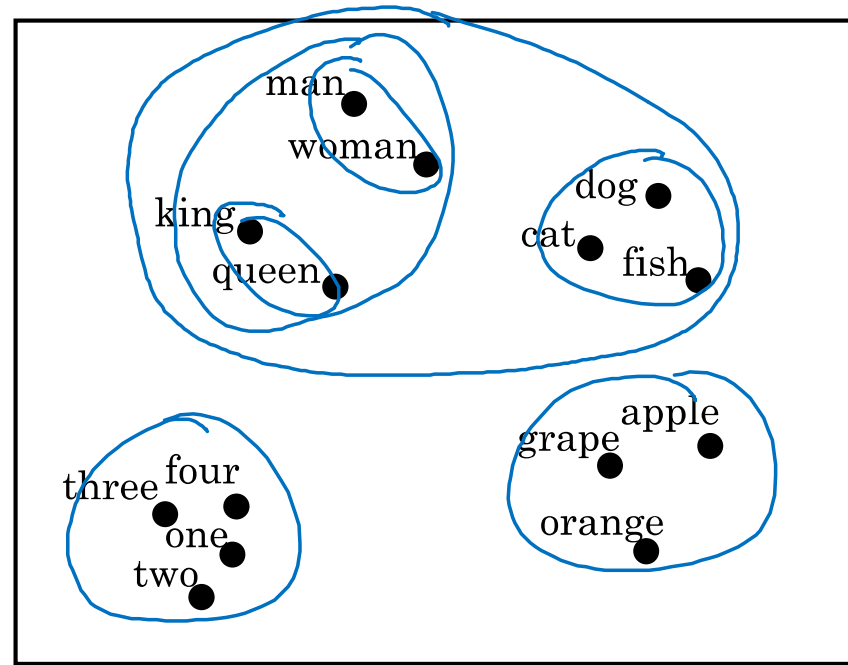
	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	Apple (456)	Orange (6257)
Gender	-1	1	-0.95	0.97	0.00	0.01
Royal	0.01	0.02	<u>0.93</u>	<u>0.95</u>	-0.01	0.00
Age	0.03	0.02	0.7	0.69	0.03	-0.02
Food	0.04	0.01	0.02	0.01	0.95	0.97
size	⋮	⋮				
cost						
alive						
verb						

I want a glass of orange juice.

I want a glass of apple juice.

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Visualizing word embeddings

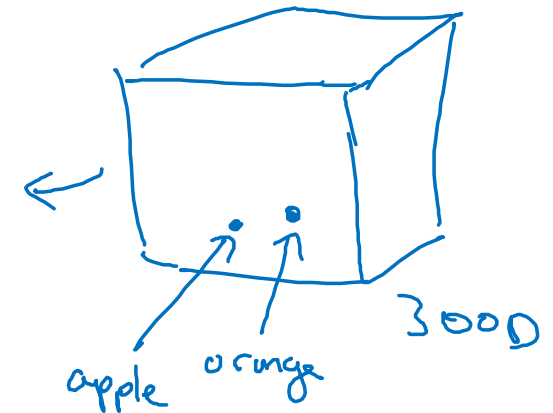


t-SNE

→ 300D



2D



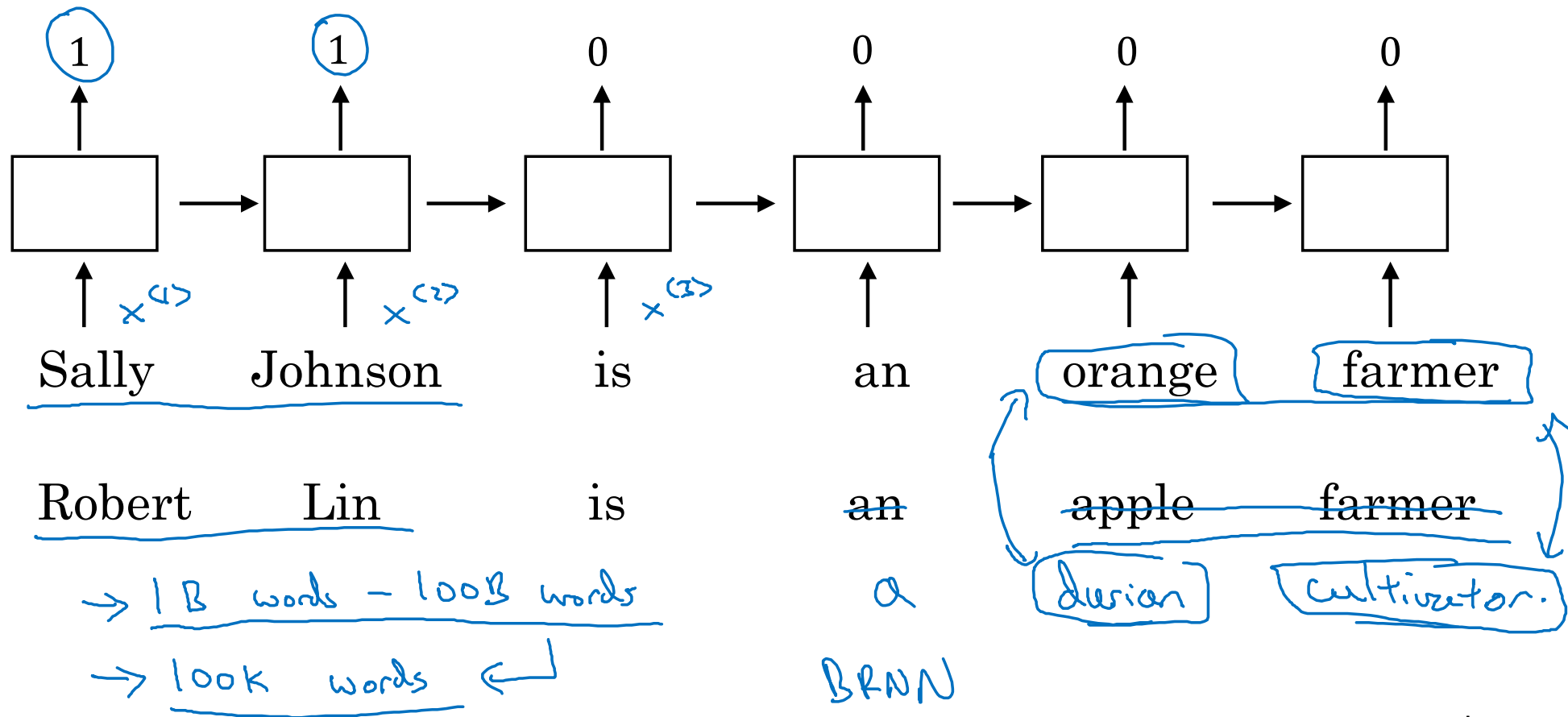


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Using word
embeddings

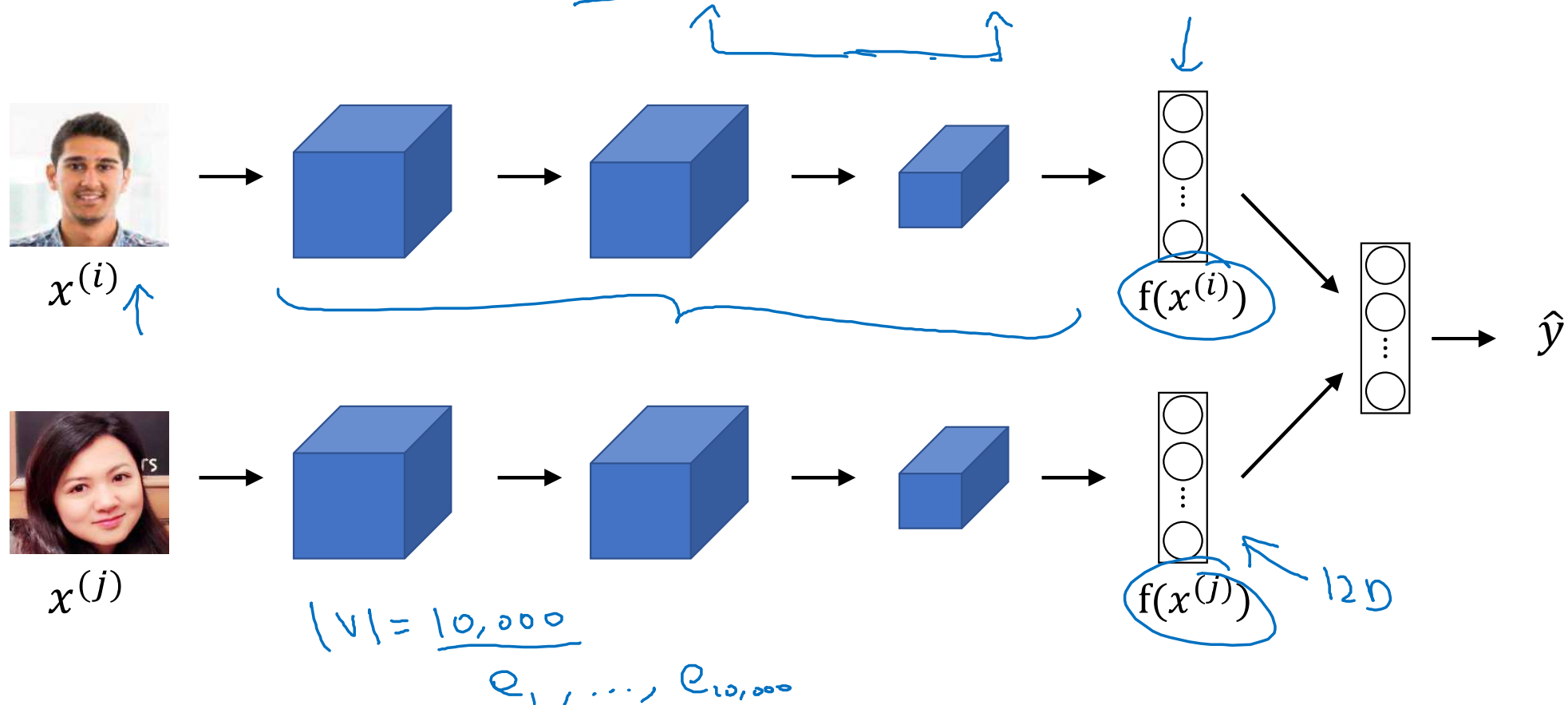
Named entity recognition example



Transfer learning and word embeddings

- 
1. Learn word embeddings from large text corpus. (1-100B words)
(Or download pre-trained embedding online.)
 2. Transfer embedding to new task with smaller training set.
(say, 100k words) → 10,000 → 300
 3. Optional: Continue to finetune the word embeddings with new data.

Relation to face encoding (embedding) 128D



[Taigman et. al., 2014. DeepFace: Closing the gap to human level performance]

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NLP and Word Embeddings

Properties of word embeddings

Analogy

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	Apple (456)	Orange (6257)
Gender	-1	1	-0.95	0.97	0.00	0.01
Royal	0.01	0.02	0.93	0.95	-0.01	0.00
Age	0.03	0.02	0.70	0.69	0.03	-0.02
Food	0.09	0.01	0.02	0.01	0.95	0.97

$$\underbrace{e_{\text{man}}}_{5391} - \underbrace{e_{\text{woman}}}_{9853} \approx \underbrace{e_{\text{king}}}_{4914} - \underbrace{e_{\text{queen}}}_{7157}$$

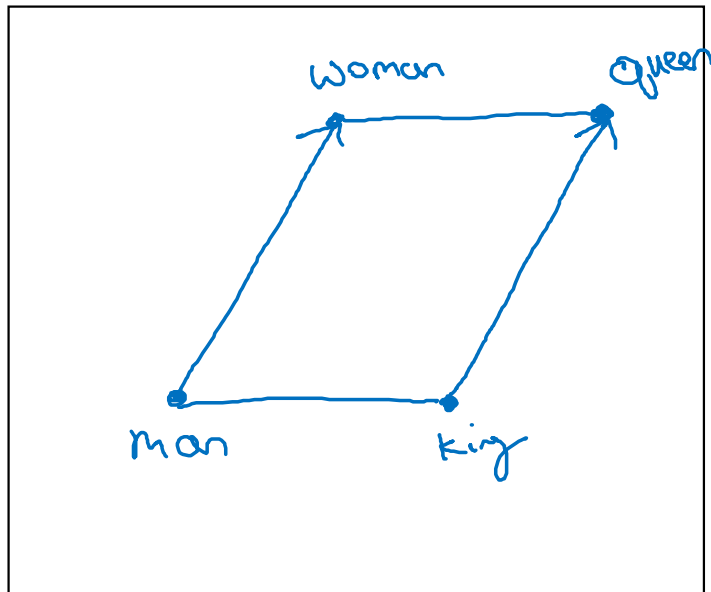
$$\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

[Mikolov et. al., 2013, Linguistic regularities in continuous space word representations]

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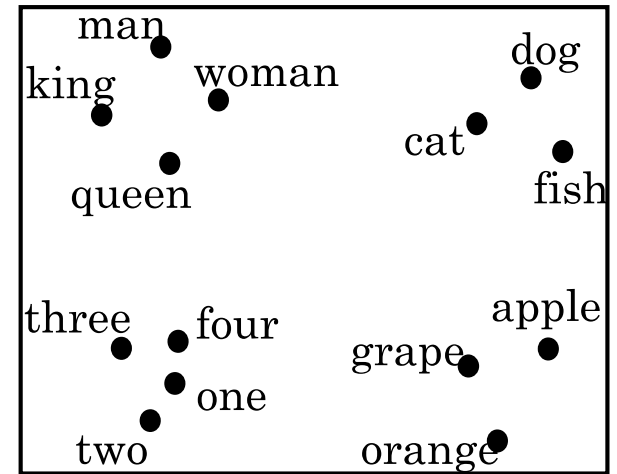
Analogies using word vectors



300 D

Find word w : $\arg \max_w$

300D $\rightarrow$ 2D
 $\uparrow$



t-SNE

$$e_{man} - e_{woman} \approx e_{king} - \cancel{e_{woman}} + e_w$$

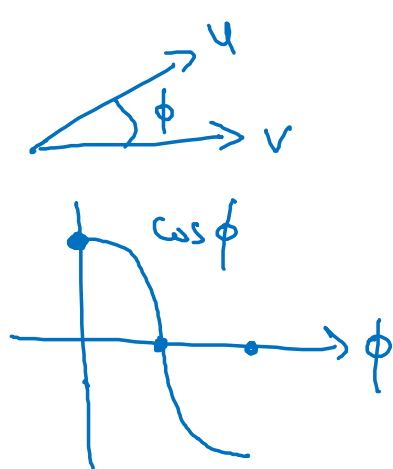
$$\text{Sim}(e_w, e_{king} - e_{man} + e_{woman})$$

30 - 75%

Cosine similarity

$$\rightarrow \boxed{\text{sim}(e_w, e_{\text{king}} - e_{\text{man}} + e_{\text{woman}})}$$

$$\text{sim}(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}$$



$$\|u - v\|^2$$

Man:Woman as Boy:Girl

Ottawa:Canada as Nairobi:Kenya

Big:Bigger as Tall:Taller

Yen:Japan as Ruble:Russia

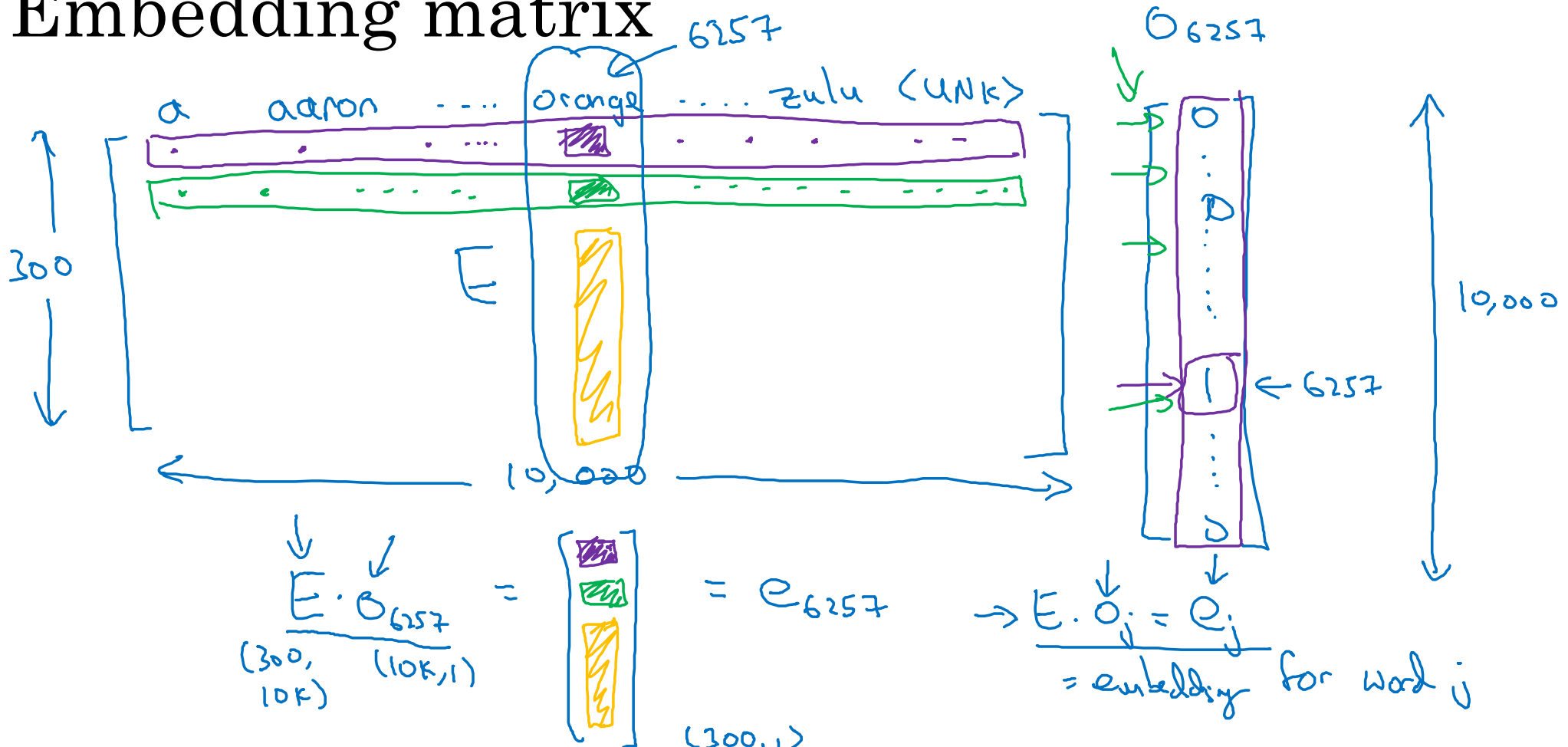


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NLP and Word Embeddings

Embedding matrix

Embedding matrix



In practice, use specialized function to look up an embedding.
 $\rightarrow \text{Embedding}$



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NLP and Word Embeddings

Learning word embeddings

Neural language model

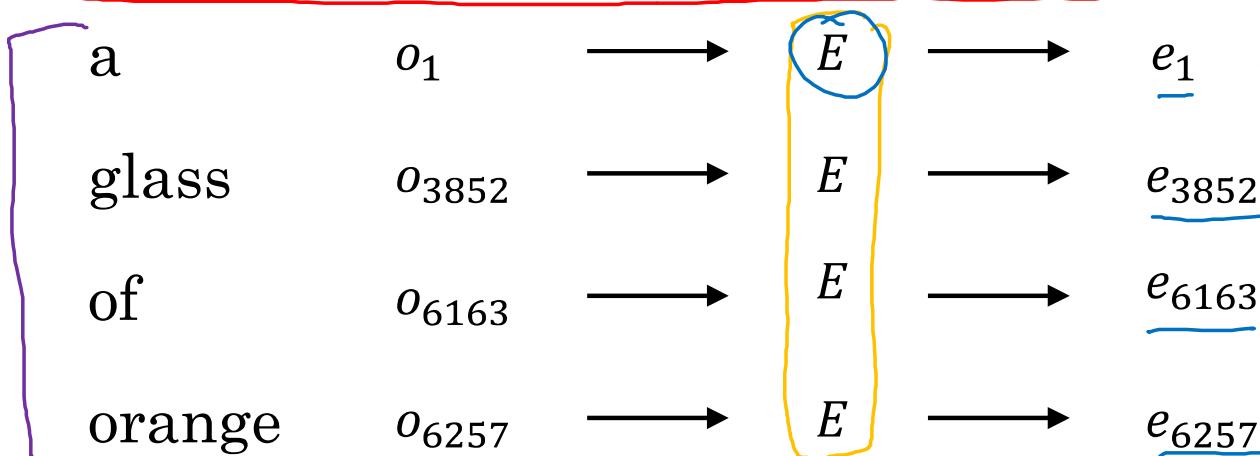
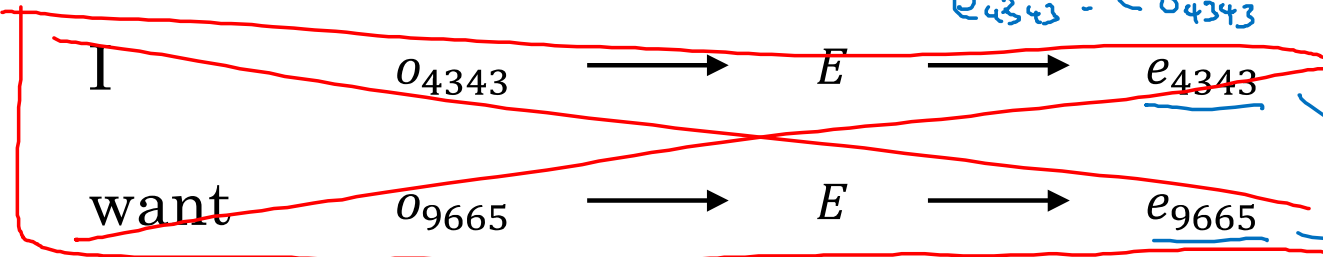
I want a glass of orange

4343 9665 1 3852 6163 6257

$$e_{4343} = E_{04343}$$

juice.

apple juice.



$$W^{(1)}, b^{(1)}$$

$$W^{(2)}, b^{(2)}$$

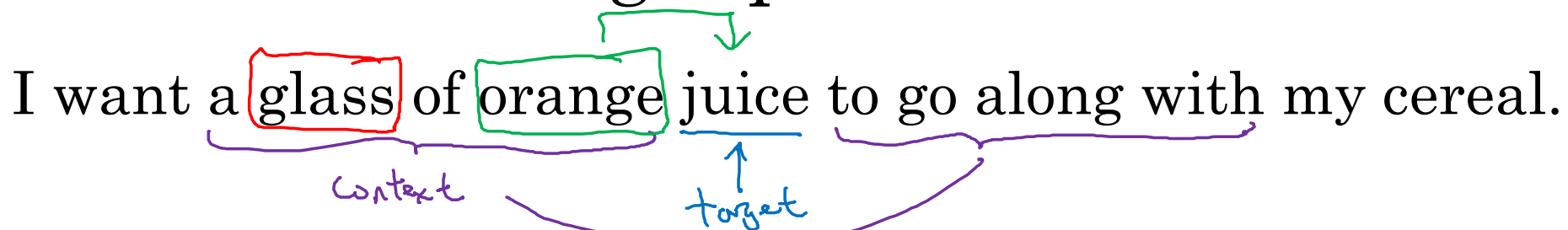
$$+1800 - 1200$$

[Bengio et. al., 2003, A neural probabilistic language model]

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Other context/target pairs

I want a glass of orange juice to go along with my cereal.



Context: Last 4 words.

- 4 words on left & right
- Last 1 word
- Nearby 1 word

a glass of orange ? to go along with

orange ?

glass ?

> skip gram



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Word2Vec

Skip-grams

I want a glass of orange juice to go along with my cereal.



Context

orange

orange

orange



Target

juice

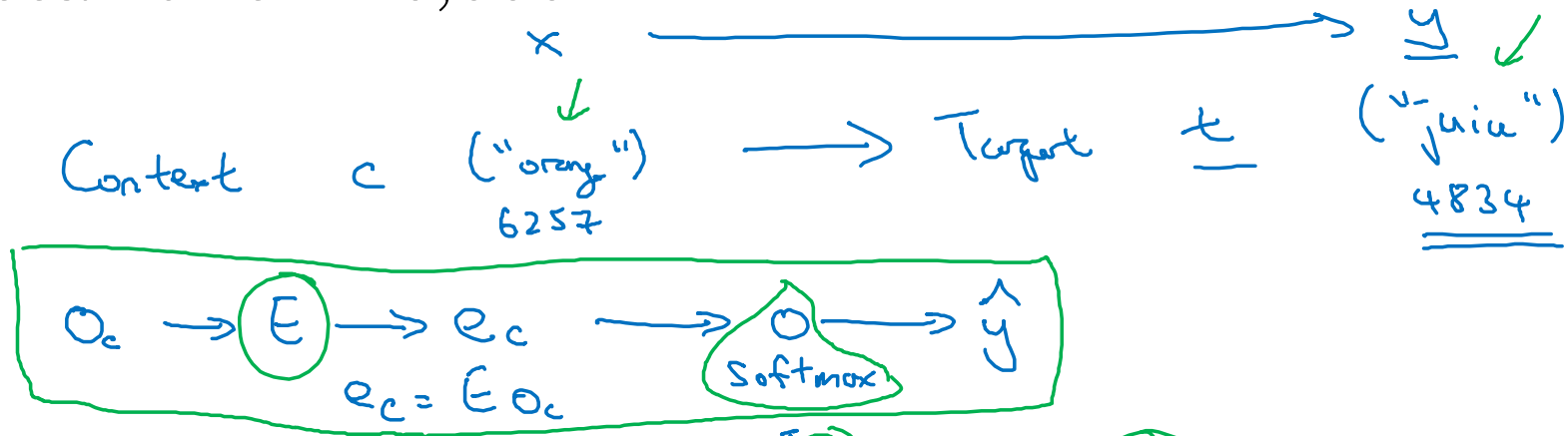
glass

my



Model

Vocab size = 10,000k



Softmax:
$$p(t|c) = \frac{e^{\theta_t^T e_c}}{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}$$

θ_t = parameter associated with output t

Loss function:

$$\mathcal{L}(\hat{y}, y) = - \sum_{i=1}^{10,000} y_i \log \hat{y}_i$$

Output vector y :

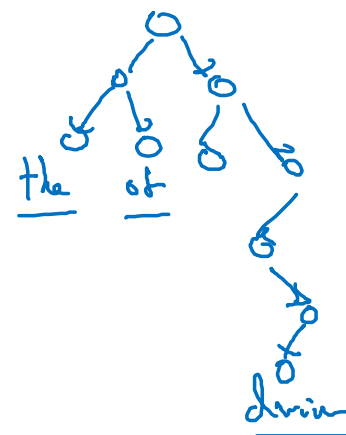
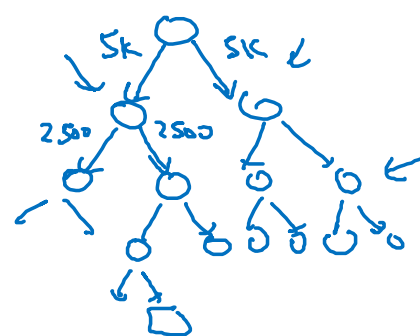
$$y = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow 4834$$

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Problems with softmax classification

$$\underline{p(t|c)} = \frac{e^{\theta_t^T \underline{e_c}}}{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}$$

Hierarchical softmax.
log |V|



How to sample the context c ?

→ the, of, a, and, to, ...

→ orange, apple, durian

P_{durian}

$P(c)$

t
 $c \rightarrow t$



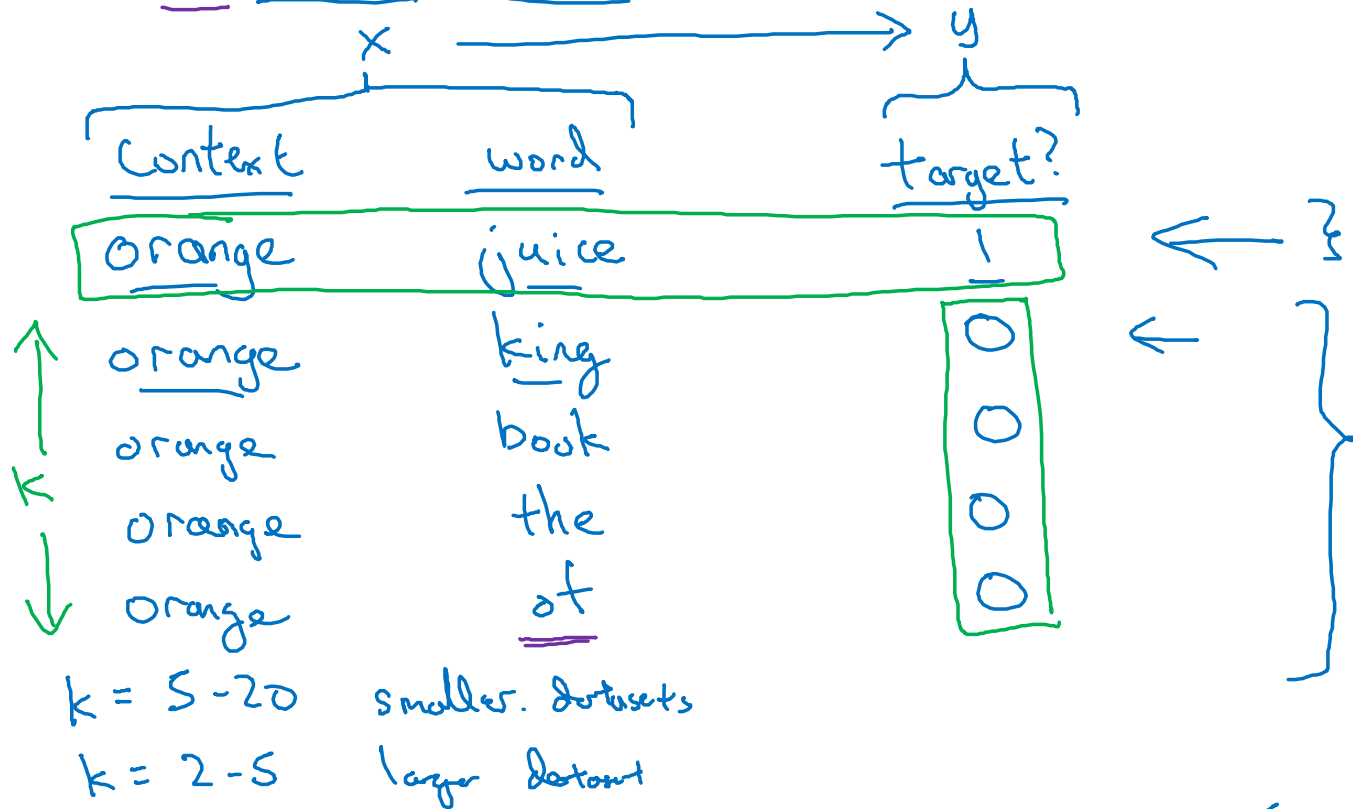
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NLP and Word Embeddings

Negative sampling

Defining a new learning problem

I want a glass of orange juice to go along with my cereal.



$k = 5-20$ smaller datasets
 $k = 2-5$ larger dataset

[Mikolov et. al., 2013. Distributed representation of words and phrases and their compositionality]

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Model

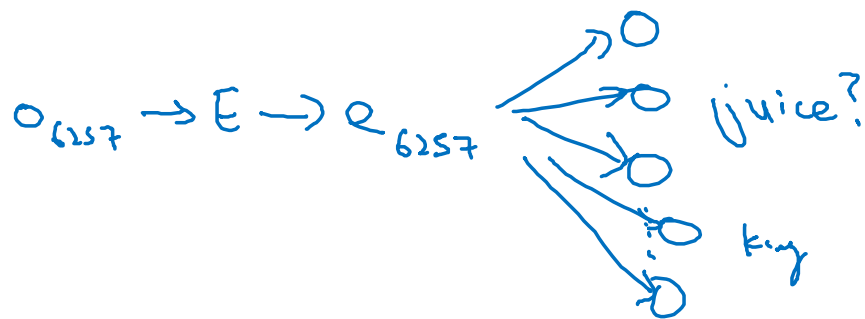
Softmax:
$$p(t|c) = \frac{e^{\theta_t^T e_c}}{\sum_{j=1}^{10,000} e^{\theta_j^T e_c}}$$

10,000-way softmax

$$P(y=1 | c, t) = \sigma(\theta_t^T e_c) \leftarrow$$

x		y
context	word	target?
orange	juice	1
orange	king	0
orange	book	0
orange	the	0
orange	of	0
c	t	y

Orange
6257



10,000 binary
classification
problem

$k+1$

Selecting negative examples

<u>context</u>	<u>word</u>	<u>target?</u>
orange	juice	1
orange	king	0
orange	book	0
orange	the	0
orange	of	0

the, of, and, ...

$$P(w_i) = \frac{f(w_i)^{3/4}}{\sum_{j=1}^{10,000} f(w_j)^{3/4}}$$

$$\frac{1}{|V|}$$

↑



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GloVe word vectors

GloVe (global vectors for word representation)

I want a glass of orange juice to go along with my cereal.

c, t

X_{ij} = # times i appears in context of j .

$\begin{matrix} \uparrow & \uparrow \\ c & t \end{matrix}$ $\begin{matrix} \uparrow \\ t \end{matrix}$ $\begin{matrix} \uparrow \\ c \end{matrix}$

$X_{ij} = X_{ji} \leftarrow$

Model

Minimize

$$\sum_{i=1}^{10,000} \sum_{j=1}^{10,000} f(x_{ij}) \left(\underbrace{\Theta_i^T e_j}_{\substack{t \quad c \\ \text{"}\Theta_t^T e_c\text{"}}} + b_i + b_j' - \log x_{ij} \right)^2$$

weighting term

$f(x_{ij}) = 0$ at $x_{ij} = 0$.

" $0 \log 0 = 0$ "

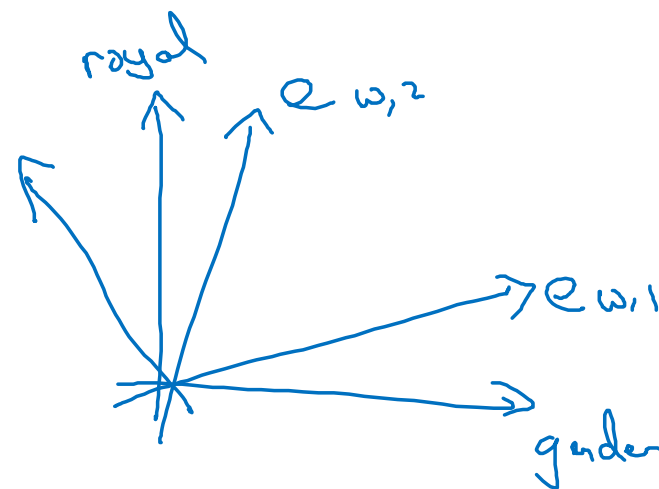
Θ_i, e_j are symmetric

$\Theta_w^{(final)} = \frac{e_w + \Theta_w}{2}$

this is, of, a,
derivation

A note on the featurization view of word embeddings

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	
Gender	-1	1	-0.95	0.97	←
Royal	0.01	0.02	0.93	0.95	←
Age	0.03	0.02	0.70	0.69	←
Food	0.09	0.01	0.02	0.01	←



$$\text{minimize } \sum_{i=1}^{10,000} \sum_{j=1}^{10,000} f(X_{ij}) (\underbrace{\theta_i^T}_{(A\theta_i)^T} e_j + b_i - b'_j - \log X_{ij})^2$$

$\hookrightarrow \underbrace{(A\theta_i)^T}_{\theta_i^T A^T} (\underbrace{e_j}_{A^T e_j}) = \theta_i^T \cancel{A^T A^T} e_j$



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NLP and Word Embeddings

Sentiment classification

Sentiment classification problem



The dessert is excellent.



Service was quite slow.



Good for a quick meal, but nothing special.



Completely lacking in good taste,
good service, and good ambience.



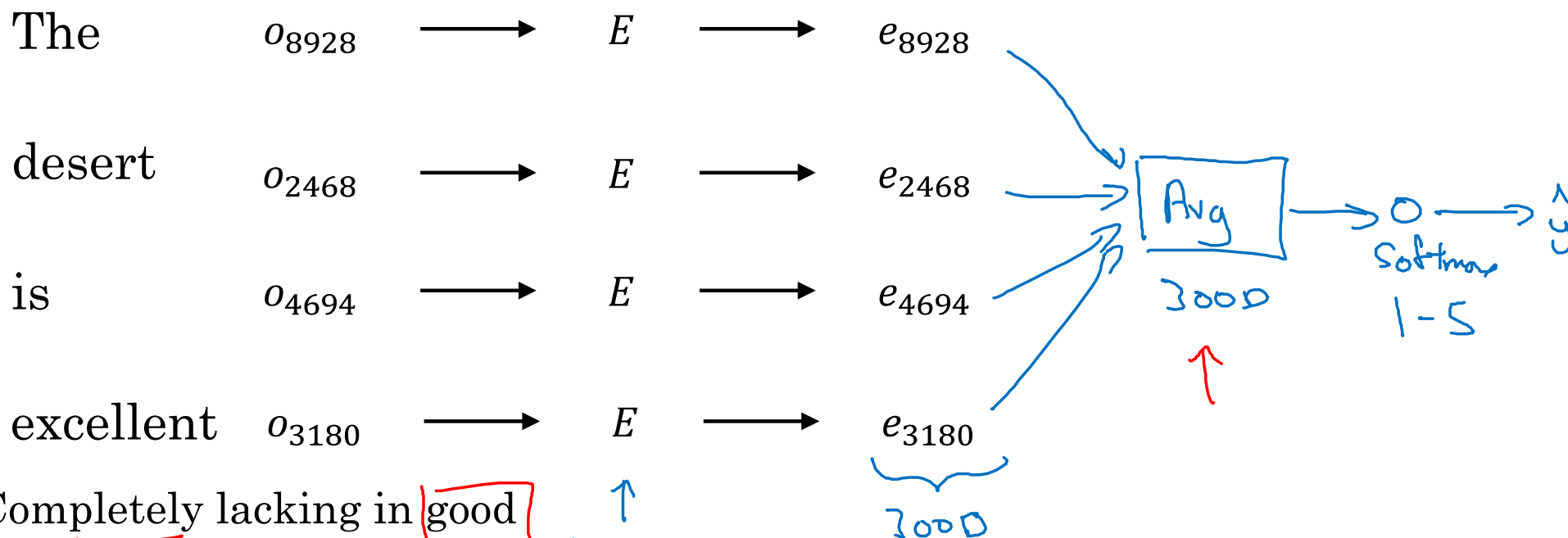
10,000 $\rightarrow$ 100,000 words

Simple sentiment classification model

The dessert is excellent



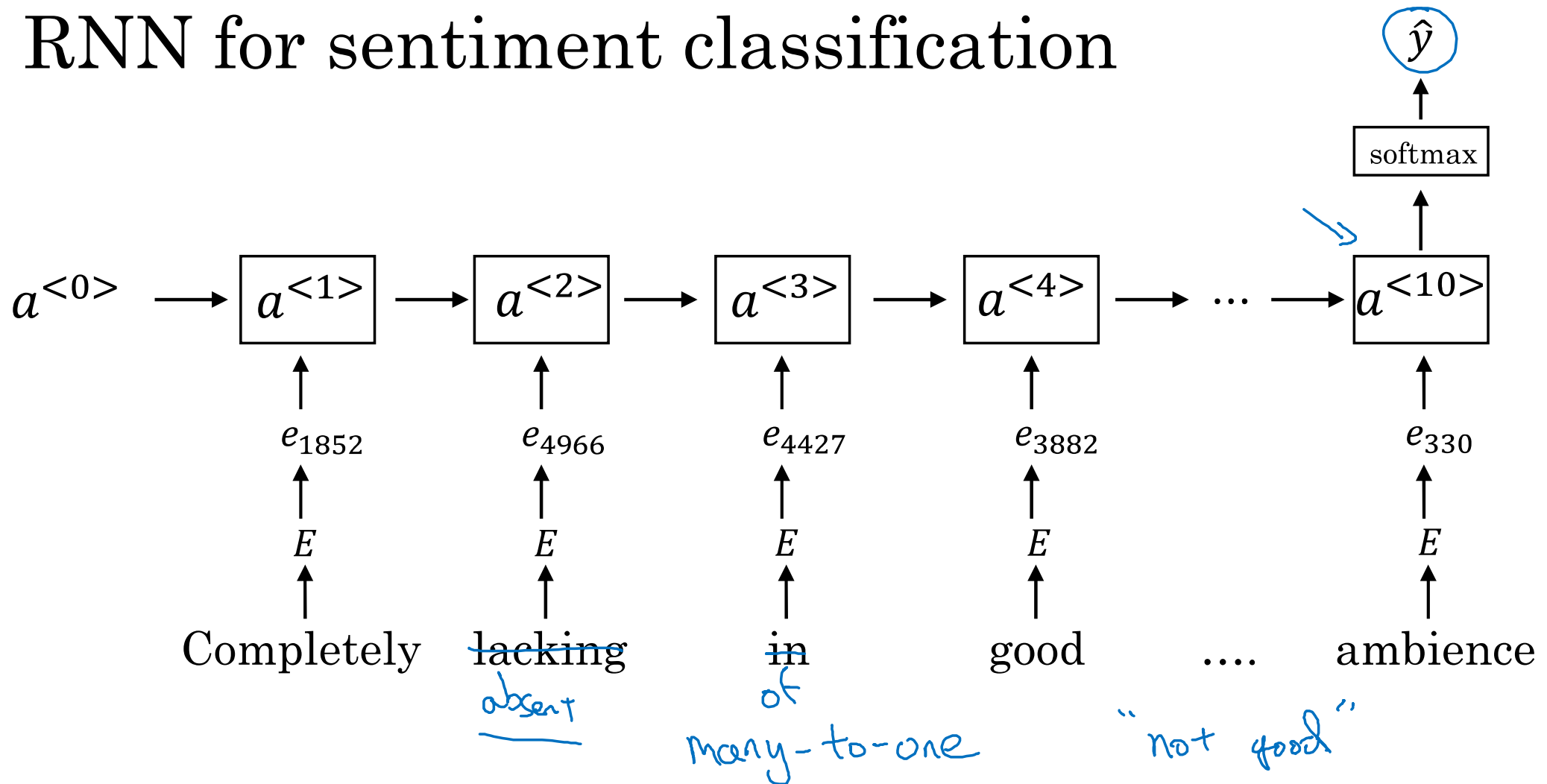
8928 2468 4694 3180



“Completely lacking in good taste, good service, and good ambience.”

↑
100 B
words

RNN for sentiment classification





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NLP and Word Embeddings

Debiasing word embeddings

The problem of bias in word embeddings

Man:Woman as King:Queen

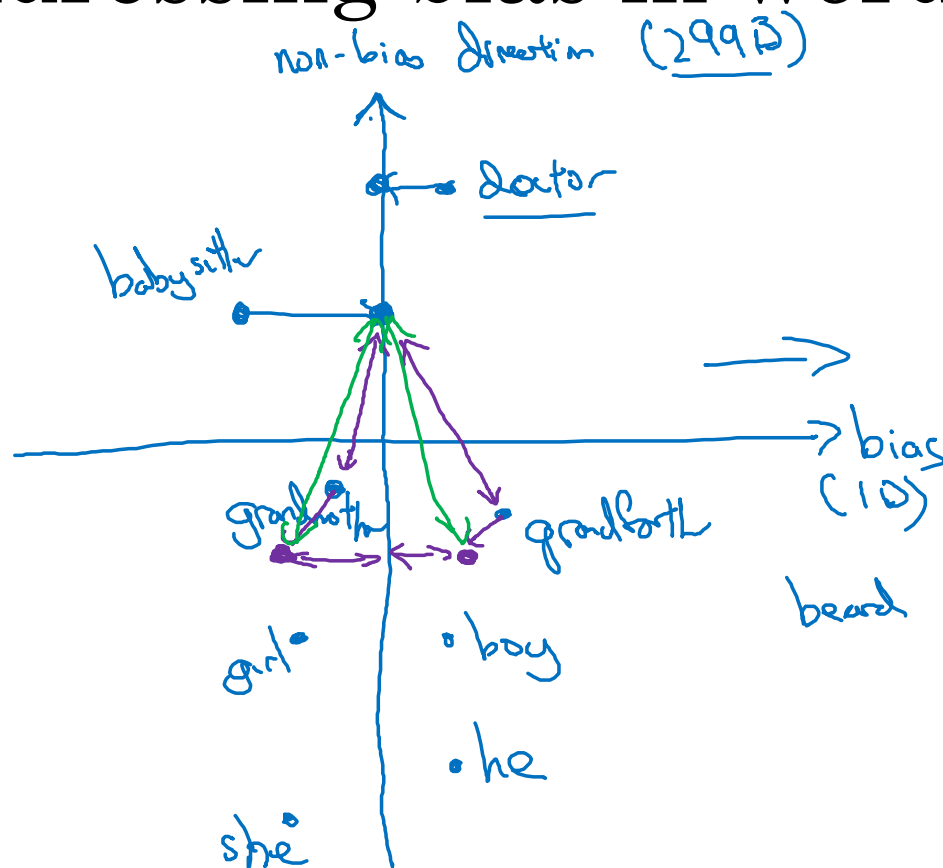
Man:Computer_Programmer as Woman:Homemaker X

Father:Doctor as Mother:Nurse X

Word embeddings can reflect gender, ethnicity, age, sexual orientation, and other biases of the text used to train the model.



Addressing bias in word embeddings



1. Identify bias direction.

$$\begin{cases} e_{he} - e_{she} \\ e_{male} - e_{female} \\ \vdots \end{cases} \rightarrow \text{average}$$

2. Neutralize: For every word that is not definitional, project to get rid of bias.

3. Equalize pairs.

$$\left\{ \begin{array}{l} \rightarrow \text{grandmother} - \text{grandfather} \\ \text{girl} \quad \text{boy} \end{array} \right\}$$